



FORM TP2013044

TEST CODE **01207010**

MAY/JUNE 2013

**CARIBBEAN EXAMINATIONS COUNCIL**  
**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®**  
**EXAMINATION**  
**BIOLOGY**

**Paper 01 - General Proficiency**

*1 hour 15 minutes*

**10 JUNE 2013 (p.m.)**

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
- (B) Diabetes
- (C) Influenza
- (D) Anaemia

Sample Answer



The best answer to this item is "Anaemia", so answer space (D) has been shaded.

5. If you want to change your answer, erase it completely and fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to this item later. Your score will be the total number of correct answers.
7. Figures are not necessarily drawn to scale.

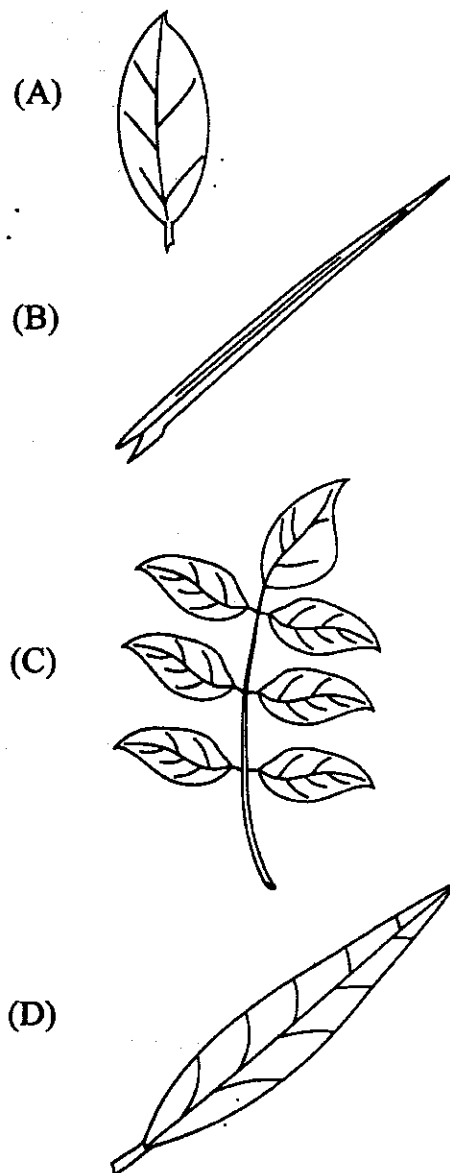
**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

1. Which of the following organisms occupies the fourth trophic level in the food chain below?

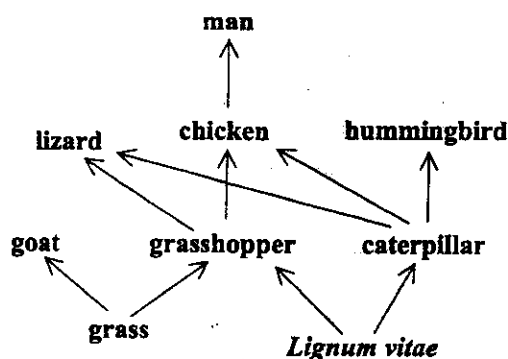
Cabbage → Caterpillar → Small bird  
→ Hawk

- (A) Hawk  
(B) Cabbage  
(C) Small bird  
(D) Caterpillar
2. Which of the following describes the relationship which involves two species such that one partner benefits while the other, though not benefiting, is unharmed?
- (A) Mutualism  
(B) Parasitism  
(C) Commensalism  
(D) Predator/Prey
3. If all the trees in a large area of a forest were 'wiped out' by disease, what would be the MOST likely effect on the composition of the air?
- (A) There would be more oxygen.  
(B) The level of carbon dioxide would decrease.  
(C) There would be no change in the composition of the air.  
(D) The level of carbon dioxide would increase.

4. Which of the following leaves does NOT have a net venation?



Item 5 refers to the following food web.



5. A herbivore, carnivore and omnivore respectively are

(A) goat, lizard, man  
(B) grass, grasshopper, lizard  
(C) caterpillar, hummingbird, goat  
(D) *Lignum vitae*, grasshopper, chicken

6. The variation in cellular structure found in any complex organism is due to a process called

(A) growth  
(B) elaboration  
(C) metabolism  
(D) specialization

7. Which of the following statements is NOT true about osmosis and diffusion?

(A) Osmosis occurs only in living cells.  
(B) Both processes require energy to move particles across a concentration gradient.  
(C) Osmosis requires a partially permeable membrane.  
(D) With both processes, molecules move from an area of high concentration to an area of lower concentration.

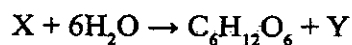
8. When compared to a cheek cell, a muscle cell contains more

(A) vacuoles  
(B) ribosomes  
(C) mitochondria  
(D) chromosomes

9. Autotrophic nutrition does NOT occur in

(A) moss  
(B) zooplankton  
(C) tomato plants  
(D) blue-green algae

10. The following chemical equation is incomplete for photosynthesis.

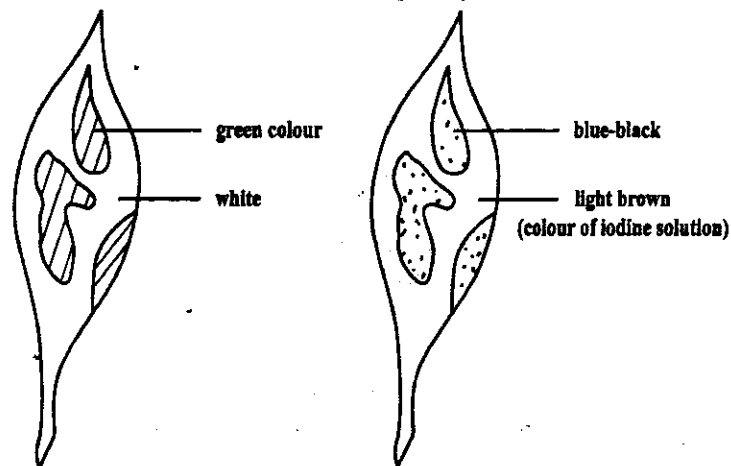


Which of the combinations shown below would accurately complete the equation?

|     | X       | Y       |
|-----|---------|---------|
| (A) | $6O_2$  | $6CO_2$ |
| (B) | $6CO_2$ | $6O_2$  |
| (C) | $O_2$   | $CO_2$  |
| (D) | $CO_2$  | $O_2$   |

Item 11 refers to the following experiment.

A green and white variegated leaf is exposed to sunlight for three hours and then tested for starch. The results of the test are shown in the following diagram.



BEFORE

AFTER

11. Based on the experiment, which of the following factors are necessary for photosynthesis?

(A) Chlorophyll and oxygen  
(B) Light and carbon dioxide  
(C) Light and chlorophyll  
(D) Chlorophyll and carbon dioxide

12. Which of the following structures found in the leaves of a flowering plant contain chloroplasts?

I. Mesophyll layer  
II. Epidermis  
III. Guard cells  
IV. Vein

(A) I and II only  
(B) I and III only  
(C) II and III only  
(D) II and IV only

Item 13 refers to the following information.

A man has poor night vision and his gums bleed whenever he brushes his teeth.

13. Which two vitamins are MOST likely lacking in his diet?

(A) A and B<sub>1</sub>  
(B) B<sub>1</sub> and B<sub>2</sub>  
(C) C and A  
(D) C and B<sub>2</sub>

14. The part of the gut into which bile empties is the

(A) ileum  
(B) caecum  
(C) stomach  
(D) duodenum

15. Which of the following is an effect of nicotine found in cigarette smoke?

(A) Increased heart rate  
(B) Increased beating of cilia  
(C) Reduced oxygen transport by the blood  
(D) Increased mucus production in the cells lining the respiratory passage

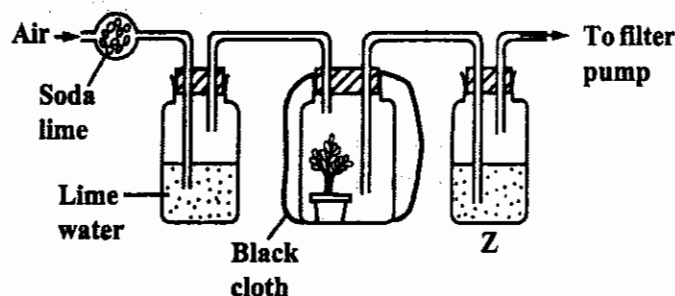
Item 16 refers to the following characteristics.

I. Thin-walled surfaces  
II. Excellent blood supply  
III. Always moist  
IV. Large surface area

16. These characteristics describe structures that are MOST likely to be associated with the

(A) diffusion of gases  
(B) production of energy  
(C) movement of water  
(D) transport of digested food

Item 17 refers to the following apparatus set up to test for products of aerobic respiration.



17. The solution in Jar Z should be

(A) iodine  
(B) tap water  
(C) calcium hydroxide (lime water)  
(D) sodium hydroxide (caustic soda)

18. The right ventricle of the heart collects blood from the

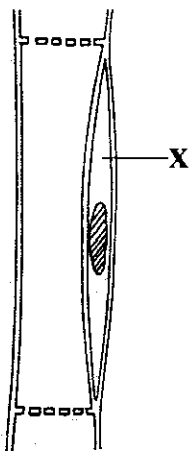
(A) aorta  
(B) vena cava  
(C) right atrium  
(D) pulmonary artery

19. Which of the following sets of characteristics is true for white blood cells?

|     | Different Cell Types | Cell Wall | Nucleus |
|-----|----------------------|-----------|---------|
| (A) | ✓                    | ✓         | ✓       |
| (B) | ✓                    | ✓         | X       |
| (C) | X                    | X         | ✓       |
| (D) | ✓                    | X         | ✓       |

✓ - present  
X - absent

Item 20 refers to the following diagram which shows cells from the stem of a plant.



20. The part labelled X is the
- (A) cork cell
  - (B) cambial cell
  - (C) cortical cell
  - (D) companion cell
21. Under which of the following conditions will the rate of transpiration in *Impatiens* plants be HIGHEST?
- (A) Sunny and low wind speed
  - (B) Sunny and high wind speed
  - (C) Cloudy and low wind speed
  - (D) Cloudy and high wind speed

22. Which of the following organisms has the LEAST need for a transport system?

- (A) *Amoeba*
- (B) Tadpole
- (C) Seedling
- (D) Honeybee

23. Which of the following storage organs characterizes the Irish potato?

- (A) Bulb
- (B) Corm
- (C) Root tuber
- (D) Stem tuber

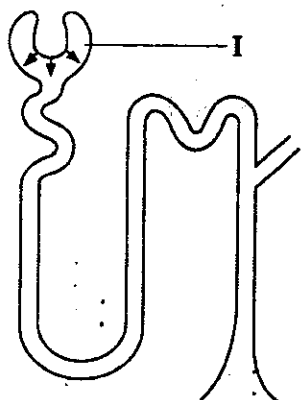
24. Excretion is important in living organisms because metabolic waste products

- (A) cannot be stored
- (B) decay very readily
- (C) are very soluble substances
- (D) can poison living organisms

25. Which of the following would NOT represent a form of excretion in plants?

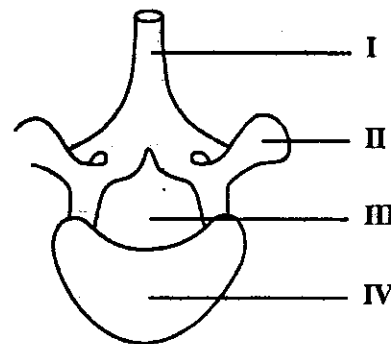
- (A) Gases diffusing through stomata
- (B) Tannins being stored in dead tissue
- (C) Water vapour diffusing through epidermis
- (D) Stored calcium oxalate crystals in leaves during leaf fall

**Item 26** refers to the following diagram of a kidney tubule.



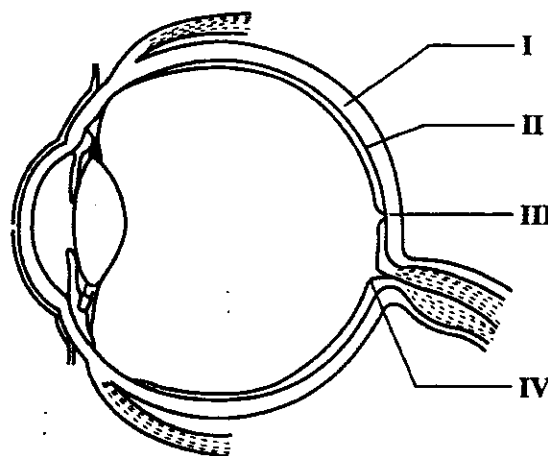
26. The glomerular filtrate in I contains
- (A) salts, water, urea, glucose
  - (B) glucose, salts, urine, water
  - (C) plasma proteins, water, urea, urine
  - (D) urine, plasma proteins, glucose, urea
27. The MAIN function of cartilage at the end of long bones is to
- (A) protect nerves
  - (B) reduce friction
  - (C) lubricate joints
  - (D) increase flexibility
28. Which of the following correctly describes movement in plants?
- (A) Irreversible, whole movement
  - (B) Reversible, whole movement
  - (C) Irreversible, growth movement
  - (D) Reversible, growth movement

**Item 29** refers to the following diagram of a thoracic vertebra.



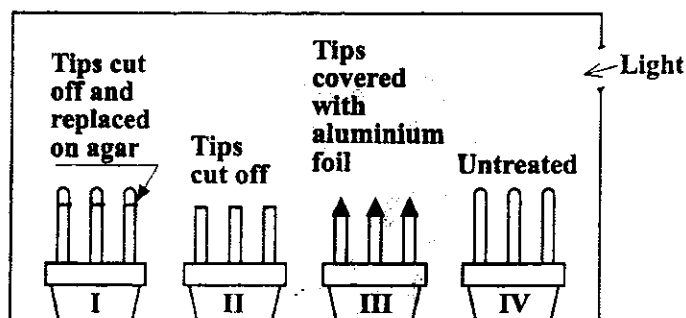
29. Nervous tissue is found at
- (A) I
  - (B) II
  - (C) III
  - (D) IV

**Item 30** refers to the following diagram of a longitudinal section through the human eye.



30. At what point is vision NOT possible?
- (A) I
  - (B) II
  - (C) III
  - (D) IV

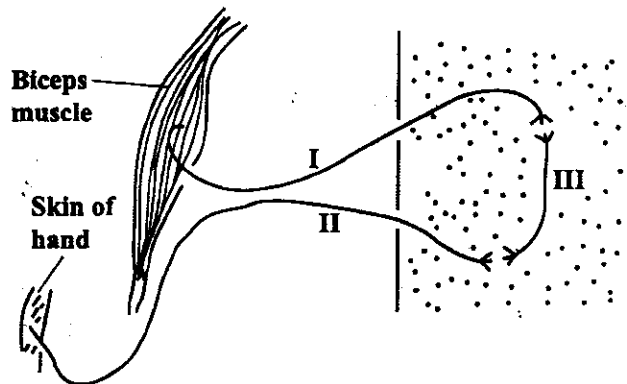
Item 31 refers to the following diagrams of germinating seedlings treated as shown.



31. At the end of one week, which seedlings would bend towards the light?

- (A) I and IV only
- (B) II and III only
- (C) I, II and III only
- (D) II, III and IV only

Item 32 refers to the following diagram.



32. The numbered structures are

- |     | I        | II            | III           |
|-----|----------|---------------|---------------|
| (A) | receptor | effector      | relay neurone |
| (B) | receptor | relay neurone | effector      |
| (C) | effector | relay neurone | receptor      |
| (D) | effector | receptor      | relay neurone |

33. Which of the following are MOST likely to cause **improper** functioning of the eye?

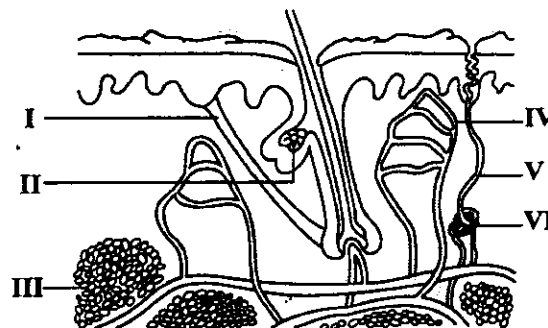
- I. Weak ciliary muscles
- II. Lack of Vitamin A
- III. Haemophilia

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

34. Which of the following endocrine glands is found in the brain?

- (A) Thyroid
- (B) Adrenal
- (C) Thymus
- (D) Pituitary

Item 35 refers to the following diagram of a vertical section through human skin.



35. Which structure(s) above will be involved in temperature regulation if there is a **DECREASE** in external temperature?

- (A) I only
- (B) I and IV only
- (C) I, II and III only
- (D) IV, V and VI only



36. A method of contraception that protects against both disease and pregnancy is the use of

(A) the pill  
(B) a condom  
(C) spermicide  
(D) an intrauterine device

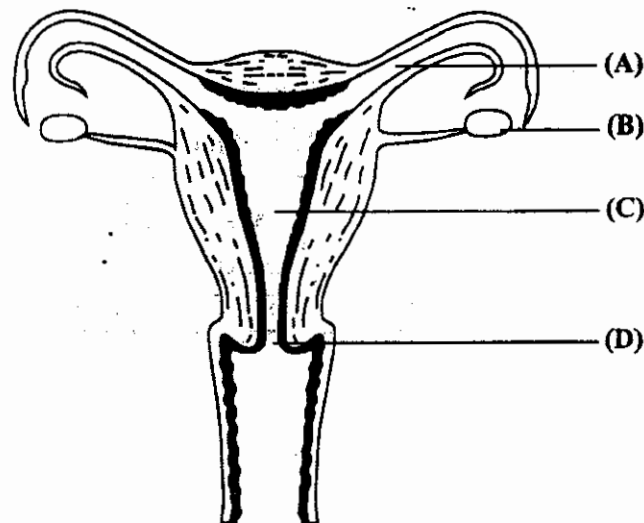
37. The endocrine gland located in the neck produces the hormone

(A) insulin  
(B) glucagon  
(C) adrenaline  
(D) thyroxine

38. Which of the following is NOT a benefit of asexual reproduction in plants?

(A) A wide variety of the offspring  
(B) Many new plants in a short time  
(C) Many plants with desirable features  
(D) New plants which are identical to each other

Items 39–40 refer to the following diagram of the female reproductive system.



In answering items 39–40, match each item with one of the options above. Each option may be used once, more than once or not at all.

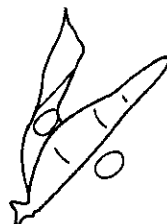
39. The structure where the egg is fertilized
40. The structure where the ova are produced
41. Seeds from the same pod were planted under the same conditions. The plants which grew from them appeared to be very similar. What is the MOST likely explanation for this occurrence?
- (A) The plant may have been self-pollinated.  
(B) The seeds had been formed asexually after fertilization.  
(C) The same pollen grain had been responsible for fertilization of the seeds.  
(D) The plant from which the seeds came became vegetatively propagated.



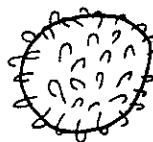
42. Which of the following fruits is dispersed by animals?



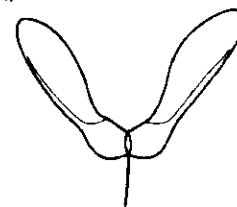
(A)



(B)



(C)



(D)

43. Which of the following are true about the production of gametes in humans?

- I. Males produce more gametes than females.
- II. Females produce more gametes than males.
- III. Males produce gametes for a longer period of their life span than females.
- IV. Females produce gametes for a longer period of their life span than males.

- (A) I and III only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

44. Which of the following statements about meiosis is NOT true?

- (A) Recombination of genes occurs.
- (B) Produces four daughter cells.
- (C) Occurs only in the sex organs.
- (D) Chromosome number is maintained.

45. An example of discontinuous variation is

- (A) intelligence
- (B) leaf size
- (C) eye colour
- (D) presence of horns in cattle

46. Which of the following terms refers to visible characteristics of living things?

- (A) Gene
- (B) Phenotype
- (C) Genotype
- (D) Chromosome

47. In guinea pigs, rough coat is dominant to smooth coat. If two heterozygous rough-coat guinea pigs mated, they would MOST likely produce

- (A) rough-coat offspring only
- (B) smooth-coat offspring only
- (C) an equal number of rough-coat and smooth-coat offspring
- (D) more rough-coat offspring than smooth-coat offspring

48. Human beings have been able to develop many species of plants and animals with beneficial characteristics. This process is called

- (A) artificial selection
- (B) natural selection
- (C) fertilization
- (D) genetics

49. In which of the following parts of a plant does the process of mitosis occur?

- (A) Phloem
- (B) Xylem
- (C) The region behind the root tip
- (D) The region behind the root hair

Items 50–51 refer to the following.

- (A) Host
- (B) Vector
- (C) Parasite
- (D) Pathogen

In answering items 50–51, match the following statements to one of the options above. Each option may be used once, more than once or not at all.

50. An organism which causes a disease

51. An animal which carries a disease-causing organism but is not adversely affected by it

52. Which of the following are involved in the formation of blood clots?

- (A) Platelets
- (B) Antibodies
- (C) Phagocytes
- (D) Lymphocytes

53. Which of the following methods may help to reduce the spread of diseases caused by insect vectors?

- I. Sterilization
- II. Use of insecticides
- III. Use of disinfectants

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II and III

54. It is believed that AIDS is NOT transmitted

- (A) from mother to foetus through the placenta
- (B) by sharing food with an infected person
- (C) by using contaminated needles
- (D) by sexual intercourse

55. You can reduce your likelihood of getting a non-communicable disease by changing risk factors that relate to your

- (A) age
- (B) heredity
- (C) lifestyle
- (D) body type

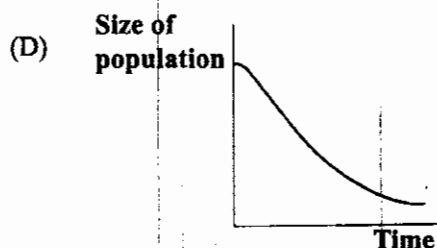
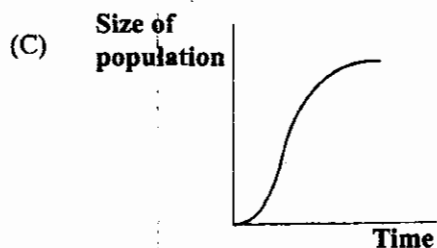
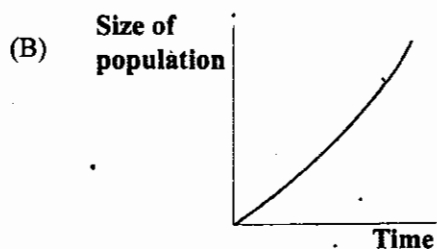
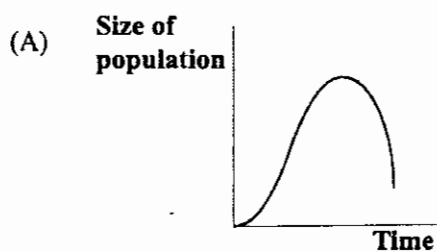
56. Which of the following provides the BEST example of an ecological community?

- (A) A colony of termites
- (B) A flock of migrating birds
- (C) Flies on a piece of rotting meat
- (D) The organisms in a freshwater pond

57. Which of the following statements is true about inorganic fertilizers?

- (A) They do not affect soil pH.
- (B) They help to hold water in the soil.
- (C) They make nutrients available more quickly.
- (D) They improve the crumb structure of soil.

Items 58–59 refer to the following graphs which show the growth pattern of several populations.



In answering items 58–59, match the following statements about the population, to one of the options above. Each option may be used once, more than once or not at all.

58. Has a good food supply and is NOT subject to environmental stress
59. Has reached its equilibrium and there is a delicate balance between the rate of birth and the rate of death

60. Which of the following organisms are MOST important in biodegradation?

- (A) Bacteria  
(B) Vultures  
(C) Omnivores  
(D) Earthworms